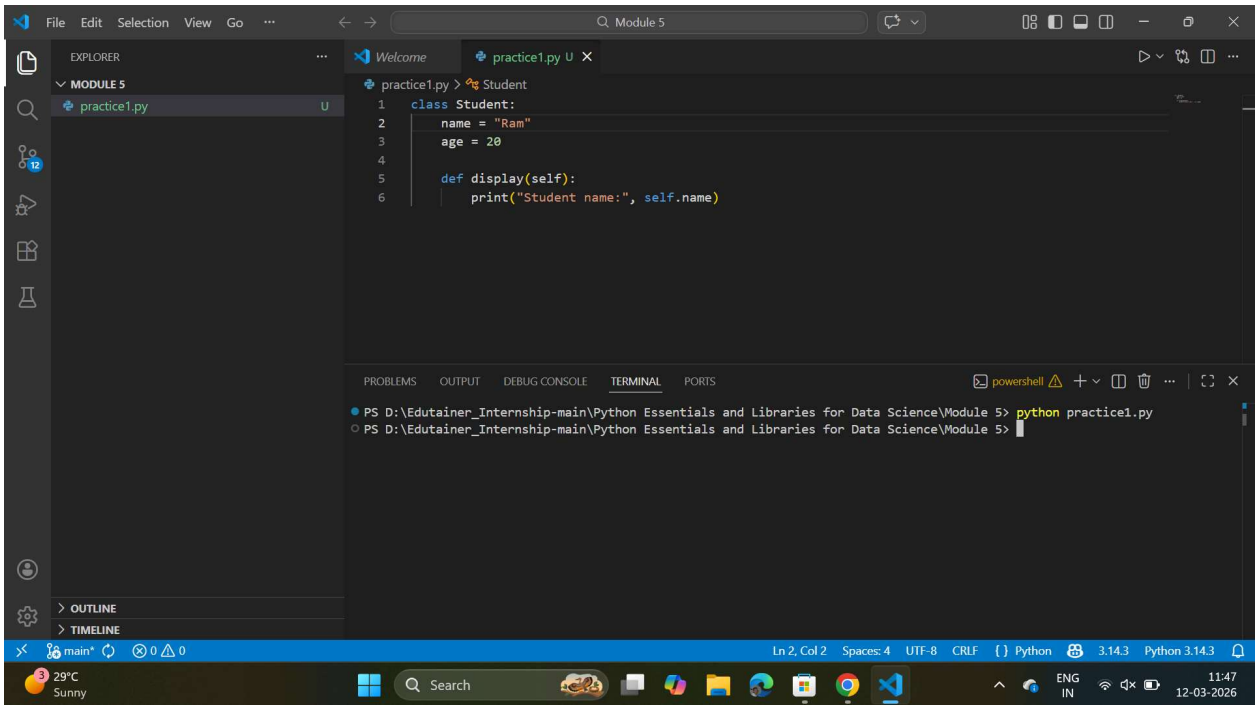
Class:-

A class is a collection of objects that share common properties and behaviour where properties are the variables that store data (also called attributes) and behaviour are the functions that define actions (also called methods).

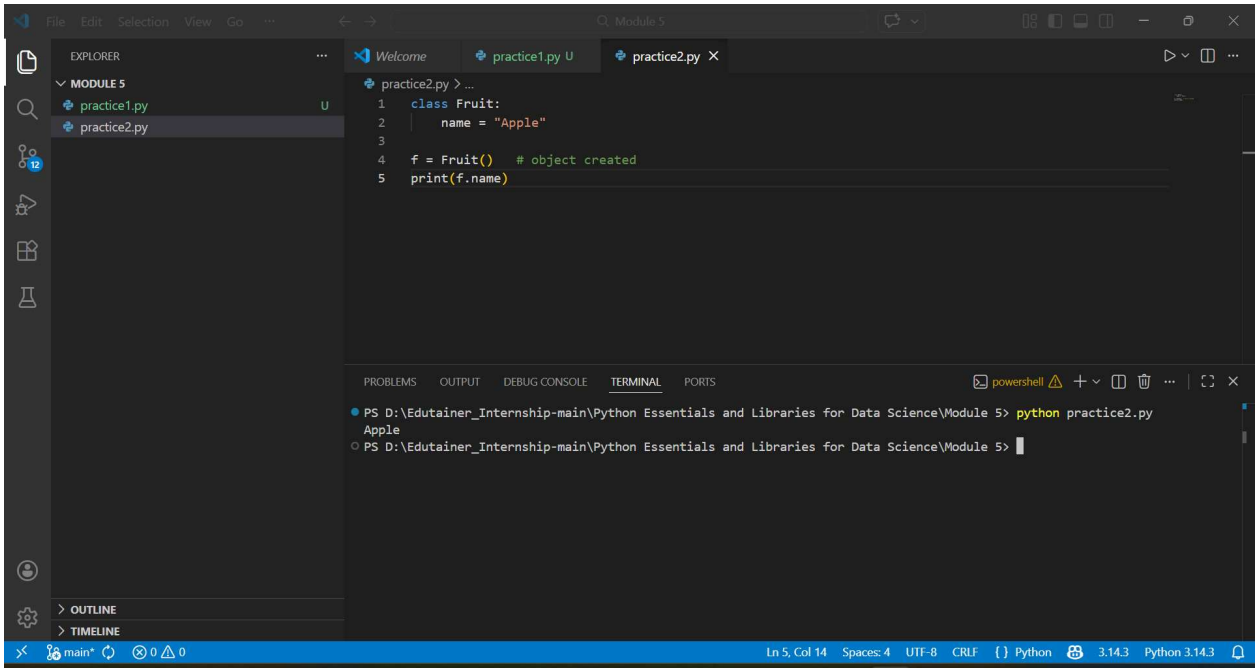


Object:-

An object is an instance of a class.A class by itself is non-existent and cannot be used.An object represents the class and provides a way to initialize values.

Example

- An apple is an object of class fruit
- A car is an object of class vehicle



Abstraction

- Is the ability of the system to only show necessary details of the system to the user.
- Hierarchical system development.

Encapsulation

- Is the ability of the system to hide unnecessary details of the system.

Inheritance

- Is the process of creating specialized classes from generalized classes.
- Also to inherit the classes provided by Python library.

Polymorphism

- A function in a class that is designed to show multiple forms.

Question

Write a Python program to capture the data regarding cake orders using a class and multiple objects.

The program should store the following details for each cake:

- Cake Name
- Shape
- Flavour

Use the following data:Create a class named Cake with appropriate attributes and a method to display the cake details.

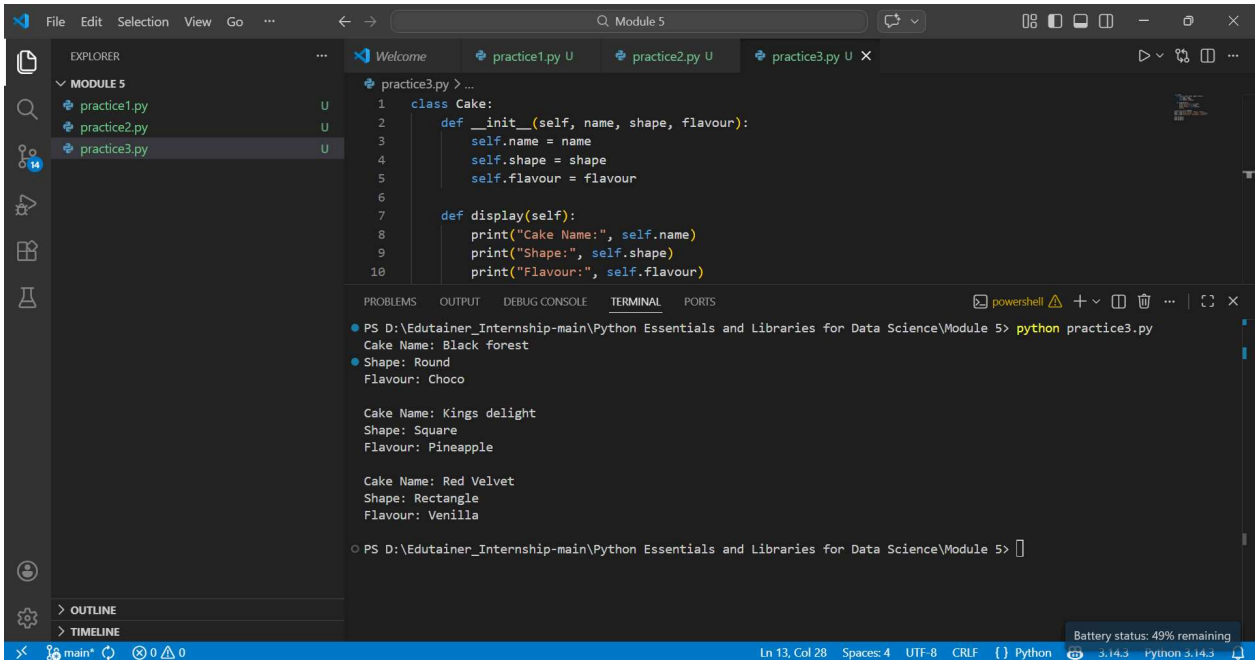
Cake Name	Shape	Flavour
Black forest	Round	Choco
Kings delight	Square	Pineapple
Red Velvet	Rectangle	Vanilla

Create multiple objects to represent the cake orders and display their information.

```
class Cake:

    def __init__(self, name, shape, flavour):

        self.name = name

        self.shape = shape

        self.flavour = flavour



    def display(self):

        print("Cake Name:", self.name)
```

```
        print("Shape:", self.shape)

        print("Flavour:", self.flavour)

        print()



# creating multiple objects

cake1 = Cake("Black forest", "Round", "Choco")

cake2 = Cake("Kings delight", "Square", "Pineapple")

cake3 = Cake("Red Velvet", "Rectangle", "Venilla")



cake1.display()

cake2.display()

cake3.display()
```

Output:-


File Edit Selection View Go
Module 5
EXPLORER
MODULE 5
practice1.py U
practice2.py U
practice3.py U
Welcome
practice1.py U
practice2.py U
practice3.py U
practice3.py > ...
class Cake:
def __init__(self, name, shape, flavour):
self.name = name
self.shape = shape
self.flavour = flavour
def display(self):
print("Cake Name:", self.name)
print("Shape:", self.shape)
print("Flavour:", self.flavour)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 5> python practice3.py
Cake Name: Black forest
Shape: Round
Flavour: Choco
Cake Name: Kings delight
Shape: Square
Flavour: Pineapple
Cake Name: Red Velvet
Shape: Rectangle
Flavour: Venilla
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 5>
OUTLINE
TIMELINE
Battery status: 49% remaining
main*
Ln 13, Col 28
Spaces: 4
UTF-8
CRLF
Python
3.14.3
Python 3.14.3